

Read operation Simplified:

$TS(T_1) \geq WTS(x) \Rightarrow$ means x was written before or at the same time T_1 started. Safe to read.

$\nexists TS(T_1) < WTS(x)$: means x was modified after T_1 started. T_1 would be reading future data. T_1 is rolled back and may be restarted with a new timestamp.

Write operation

if $TS(T_1) < RTS(x)$
 $\Rightarrow$ someone read x after T_1 started. If we let T_1 write now, it would change a value that has already been read by a newer transaction, which could cause inconsistency. T_1 is aborted.

if $TS(T_1) < WTS(x) \Rightarrow$ means x was already written by a newer transaction. T_1 is trying to write an outdated value, so it gets aborted.

$\Rightarrow$ if both checks Pass: Sails.

write is allowed and $WTS(x) \rightarrow$ updated to $TS(T_1)$

S	10
T_1	T_2
$w(x)$	
	$R(x)$
$w(x)$	

$x \rightarrow$ data object

$$WTS(x) = 0$$

$$RTS(x) = 0$$

} initial timestamp of x

$$WTS(x) \rightarrow 5$$

$$RTS(x) \rightarrow 0$$

} after 1st execution

$$WTS(x) \rightarrow 5$$

$$RTS(x) \rightarrow 10$$

} after 2nd execution

1) T_1 write x

1) $5 < 0$ false.

$5 < 0$ false.

So write ~~perform~~ executes.

← checks write rules

2) T_2 Read x

$10 < 5$ false.

$10 \geq 5$ true.

Execution happens.

← checking read rules

$$RTS(Q) = \max(RTS(Q), TS(T_2))$$

$$RTS(Q) = 10$$

← checking write rule

3) T_1 write x : $5 < 10 \rightarrow$ ~~false~~ true. so write not allowed.
 $5 < 5 \rightarrow$ ~~false~~ so roll back.